

✓ Statistical Foundations of Data Sciences — Practical 3

Student Rating / TeachingRatings Dataset

Dataset: TeachingRatings (University of Texas at Austin, 2000–2002)

This practical analyzes instructor characteristics, tenure status, age, gender, minority status, and teaching evaluation scores.

Variables used

- `minority` — whether the instructor belongs to a visible/non-Caucasian minority group (`yes`/`no`).
- `tenure` — whether the instructor is on the tenure track (`yes`/`no`).
- `age` — instructor age in years.
- `gender` — instructor gender (`male`/`female`).
- `eval` — overall teaching evaluation score.

Note: In the original dataset documentation, `tenure` refers to being **on the tenure track**, so the term “tenured” in the questions is interpreted according to the dataset variable.

✓ 1. Import Required Libraries

We use **Pandas** for data handling, **NumPy** for numerical operations, **Matplotlib** for graphs, and SciPy's `chi2_contingency` for the Chi-square test.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import chi2_contingency
```

✓ 2. Load the Dataset

Update the file path if your `TeachingRatings.csv` is stored somewhere else in Google Drive.

```
from google.colab import drive
drive.mount('/content/drive')

# Load the dataset
df = pd.read_csv('/content/drive/MyDrive/TeachingRatings.csv')

# Display the first five rows
df.head()
```

Mounted at /content/drive

	rownames	minority	age	gender	credits	beauty	eval	division	native	tenure	students	allst
0	1	yes	36	female	more	0.289916	4.3	upper	yes	yes	24	
1	2	no	59	male	more	-0.737732	4.5	upper	yes	yes	17	
2	3	no	51	male	more	-0.571984	3.7	upper	yes	yes	55	
3	4	no	40	female	more	-0.677963	4.3	upper	yes	yes	40	
4	5	no	31	female	more	1.509794	4.4	upper	yes	yes	42	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/

3. Basic Dataset Check

`df.info()` checks the number of observations, columns, missing values, and data types.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 463 entries, 0 to 462
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   rownames        463 non-null   int64
1   minority        463 non-null   object
2   age             463 non-null   int64
3   gender          463 non-null   object
4   credits         463 non-null   object
5   beauty          463 non-null   float64
6   eval            463 non-null   float64
7   division        463 non-null   object
8   native          463 non-null   object
9   tenure          463 non-null   object
10  students        463 non-null   int64
11  allstudents     463 non-null   int64
12  prof            463 non-null   int64
dtypes: float64(2), int64(5), object(6)
memory usage: 47.2+ KB
```

Q1. Percentage of Visible Minorities Who Are Tenure-Track Professors

Question

Calculate the percentage of visible minorities who are tenure-track professors. Then determine whether tenure status differs by minority status.

Step 1 — Cross-tabulation

`pd.crosstab()` creates a frequency table between `minority` and `tenure`.

`normalize='index'` converts each row into proportions. Multiplying by 100 converts them into percentages. Therefore, the `yes` value in the `minority = yes` row is the percentage of visible-minority instructors who are tenure-track.

```
minority_tenure_percent = pd.crosstab(
    df['minority'],
    df['tenure'],
    normalize='index'
) * 100

print('Tenure status by minority status (%)')
display(minority_tenure_percent.round(2))

# Percentage of visible-minority instructors who are on the tenure track
minority_tenured_percentage = minority_tenure_percent.loc['yes', 'yes']
print(f'Percentage of visible-minority instructors who are tenure-track: {minority_tenured_percentage}
```

Tenure status by minority status (%)

	tenure	no	yes
minority			
no		23.06	76.94
yes		15.62	84.38

Percentage of visible-minority instructors who are tenure-track: 84.38%

✓ Step 2 — Chi-square test of independence

The Chi-square test checks whether `minority` and `tenure` are statistically associated.

Null hypothesis (H_0): Minority status and tenure status are independent.

Alternative hypothesis (H_1): Minority status and tenure status are associated.

At the 5% significance level:

- If `p < 0.05`, reject $H_0 \rightarrow$ statistically significant association.
- If `p >= 0.05`, fail to reject $H_0 \rightarrow$ insufficient evidence of an association.

```
table = pd.crosstab(df['minority'], df['tenure'])

chi2, p_value, dof, expected = chi2_contingency(table)

print('Observed frequency table:')
display(table)
print(f'Chi-square statistic: {chi2:.4f}')
print(f'p-value: {p_value:.4f}')
print(f'Degrees of freedom: {dof}')

if p_value < 0.05:
    print('Conclusion: Tenure status differs significantly by minority status.')
else:
    print('Conclusion: There is insufficient evidence to say that tenure status differs by minority status')
```

Observed frequency table:

tenure	no	yes
minority		
no	92	307
yes	10	54

Chi-square statistic: 1.3675

p-value: 0.2422

Degrees of freedom: 1

Conclusion: There is insufficient evidence to say that tenure status differs by minority status.

Next steps: [Generate code with table](#) [New interactive sheet](#)

Q2. Does Average Age Differ by Tenure?

We need the **mean** and **standard deviation** of `age` separately for tenure-track and non-tenure-track instructors.

`groupby('tenure')` creates separate groups, and `.agg(['mean', 'std'])` calculates the mean and standard deviation for each group.

```
age_by_tenure = df.groupby('tenure')['age'].agg(['mean', 'std'])
age_by_tenure.columns = ['Mean Age', 'Standard Deviation']
display(age_by_tenure.round(3))
```

	Mean Age	Standard Deviation
tenure		
no	50.186	6.946
yes	47.850	10.420

Interpretation

Compare the two group means to see which tenure group is older on average. The standard deviation shows how much age varies within each group.

Expected descriptive results for this dataset:

- Non-tenure-track (`no`) mean age $\approx$ **50.19 years**.
- Tenure-track (`yes`) mean age $\approx$ **47.85 years**.

Thus, the non-tenure-track group is approximately 2.34 years older on average. These are descriptive differences; statistical significance requires a formal hypothesis test.

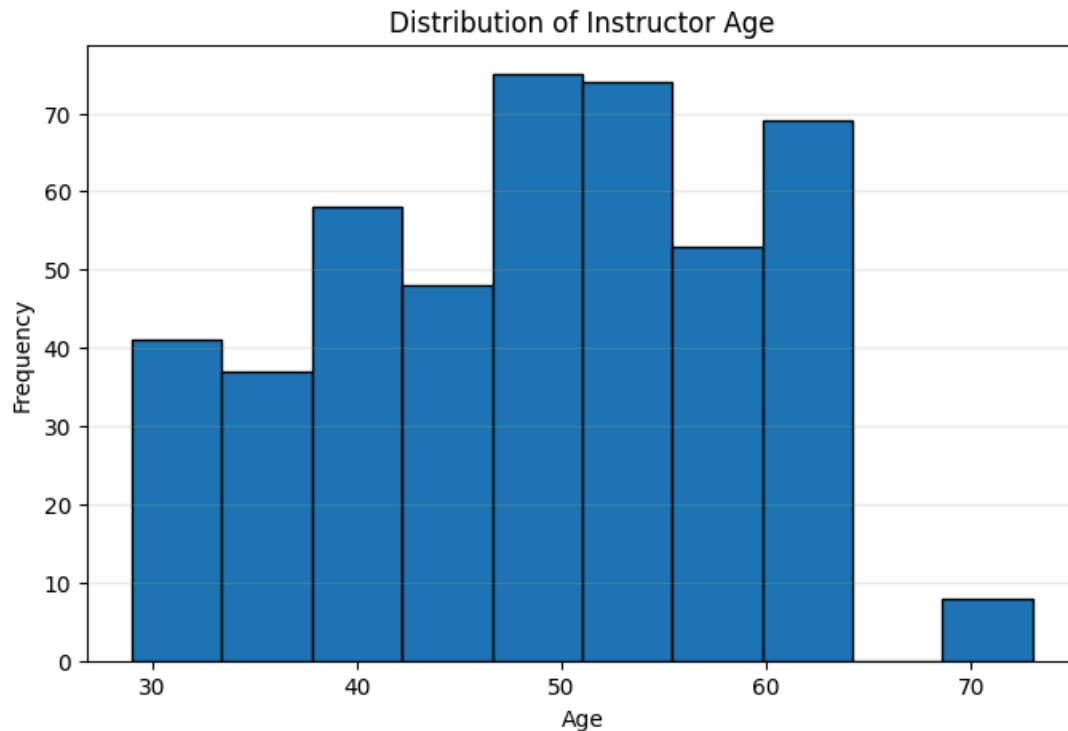
Q3. Which Graph Works Better for the Age Variable?

`age` is a **quantitative/numerical variable**. A **histogram** is appropriate because it groups ages into intervals (bins) and shows the frequency/distribution of observations.

A bar chart is generally more appropriate for categorical variables such as gender or tenure status.

```
plt.figure(figsize=(8, 5))
```

```
plt.hist(df['age'], bins=10, edgecolor='black')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Distribution of Instructor Age')
plt.grid(axis='y', alpha=0.2)
plt.show()
```



Comment

The histogram shows how instructor ages are distributed across age intervals. Look for the center of the distribution, its spread, and whether it appears symmetric or skewed.

✓ Q4. Difference Between `pyplot.bar` and `pyplot.barh`

`plt.bar()`

`plt.bar()` creates a **vertical bar chart**. Categories are placed on the x-axis and bar heights represent the values.

`plt.barh()`

`plt.barh()` creates a **horizontal bar chart**. Categories are placed on the y-axis and bar lengths represent the values.

Function	Orientation	Category axis
<code>plt.bar()</code>	Vertical	x-axis
<code>plt.barh()</code>	Horizontal	y-axis

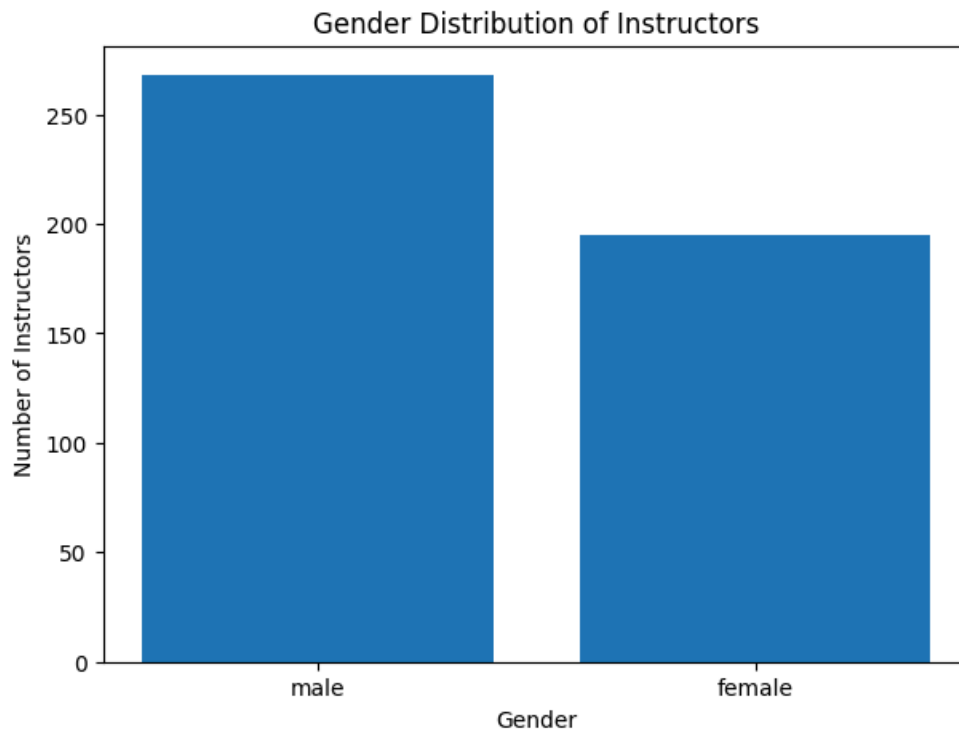
For the gender variable, either is valid. Here we use `plt.bar()` to make a simple vertical bar chart.

```
gender_counts = df['gender'].value_counts()

plt.figure(figsize=(7, 5))
```

```
plt.bar(gender_counts.index, gender_counts.values)
plt.xlabel('Gender')
plt.ylabel('Number of Instructors')
plt.title('Gender Distribution of Instructors')
plt.show()

print('Gender counts:')
display(gender_counts)
```



Gender counts:

	count
gender	
male	268
female	195

dtype: int64

✓ Q5. Median Evaluation Score for Tenure-Track Professors

We first filter the dataset to keep only instructors with `tenure == 'yes'`. Then we select the `eval` column and calculate its median.

```
tenured_eval_median = df[df['tenure'] == 'yes']['eval'].median()

print(f'Median evaluation score for tenure-track professors: {tenured_eval_median:.2f}')
```

Median evaluation score for tenure-track professors: 4.00

Interpretation

For this dataset, the median evaluation score for tenure-track instructors is **4.00**.

This means that the middle evaluation score among the tenure-track instructors is 4.00.

Practical 3 — Summary

Methods used

1. **Crosstab + row normalization** — percentage of tenure status within minority groups.
2. **Chi-square test of independence** — relationship between minority and tenure status.
3. **Group mean and standard deviation** — age comparison by tenure.
4. **Histogram** — distribution of the numerical age variable.
5. **Bar chart** — frequency of the categorical gender variable.
6. **Median** — central evaluation score among tenure-track instructors.

Important statistical rule

For hypothesis tests, use a 5% significance level unless your professor specifies another level:

α = 0.05